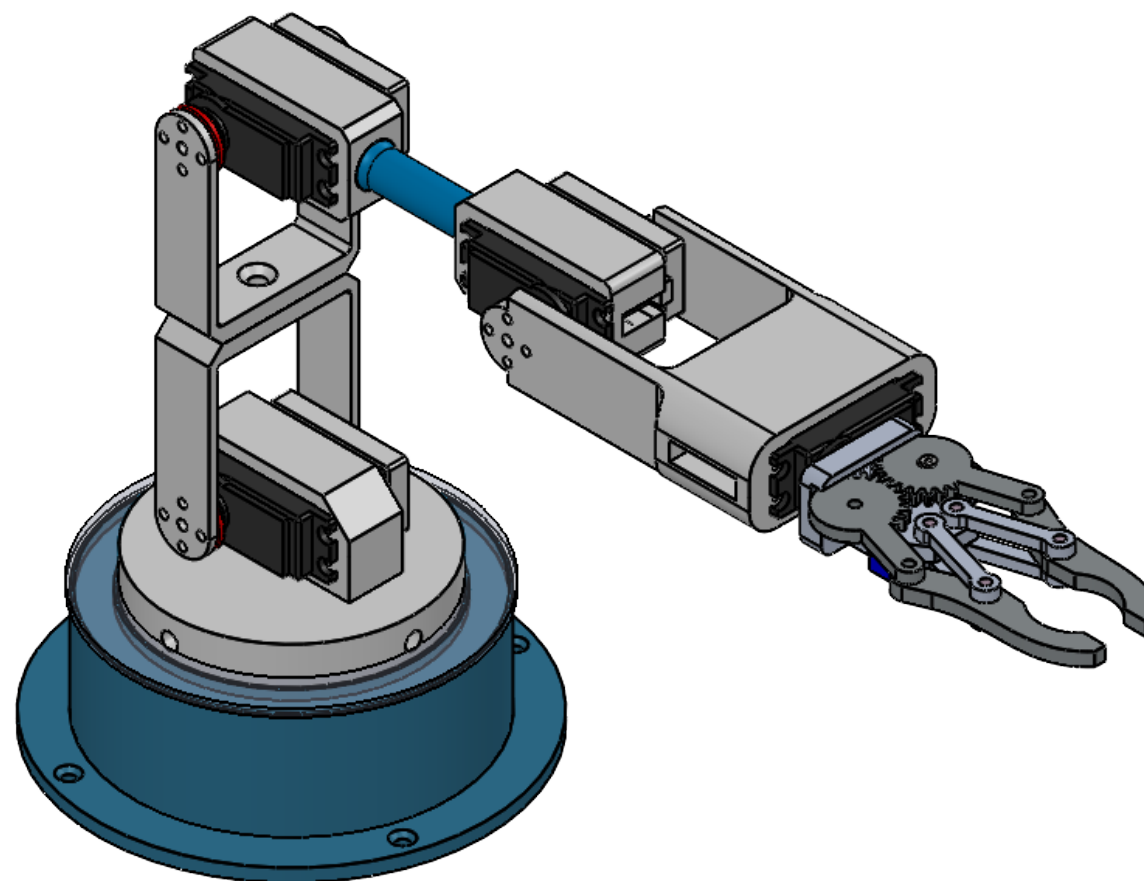
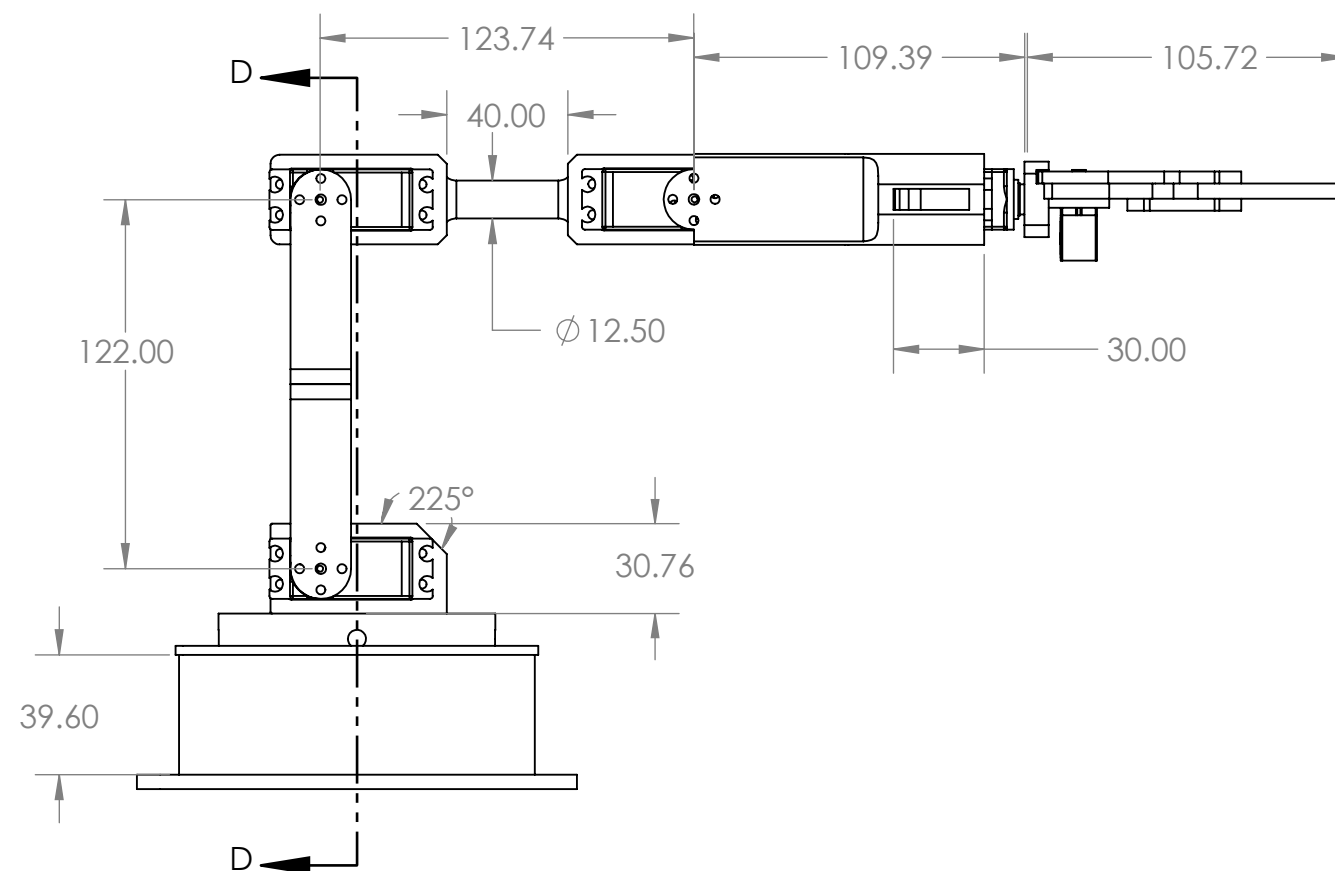
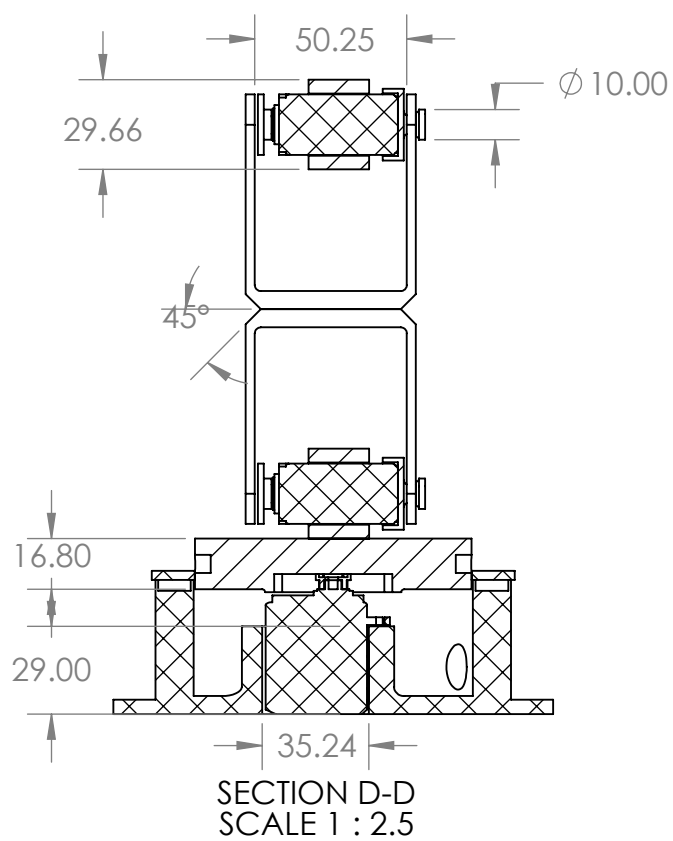




REVISION: 3

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**6-DOF Robotic
Arm**

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WEIGHT: 802.46

DWG NO.

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SHEET 1 OF 1

Design, Development, and Control for a 6-DOF Articulated Robotic Arm

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Design, Development, and Control for a 6-DOF Articulated Robotic Arm

Abstract — This paper presents the design, development, and control of a 6-DOF articulated robotic arm for precise manipulation tasks. The main objective is to develop a cost-effective and reliable robotic system capable of performing accurate position and orientation control in a constrained workspace. The system integrates a mechanical arm structure with six degrees of freedom, enabling full spatial motion and flexible end-effector positioning. A control framework is implemented to ensure smooth trajectory tracking and stable operation under varying load conditions. The main contributions of this work are: (i) the mechanical design and assembly of a 6-DOF robotic arm optimized for stability and precision, (ii) the implementation of a kinematics-based control system for accurate joint and end-effector positioning, and (iii) the validation of the system through trajectory tracking and basic manipulation tasks. Experimental results demonstrate that the robotic arm achieves consistent and accurate motion control, successfully following predefined trajectories with minimal error, validating the effectiveness of the proposed

I. Introduction

Six-degree-of-freedom robotic manipulators are widely used in industrial and research applications due to their ability to provide full spatial control for precise manipulation tasks such as assembly and pick-and-place operations [1].

However, achieving accurate and robust control remains challenging due to nonlinear dynamics, model uncertainties, and external disturbances that affect trajectory tracking performance [2]. Motivated by these challenges, this paper presents the design and kinematics-based control of a 6-DOF robotic arm aimed at improving trajectory accuracy and real-time stability.

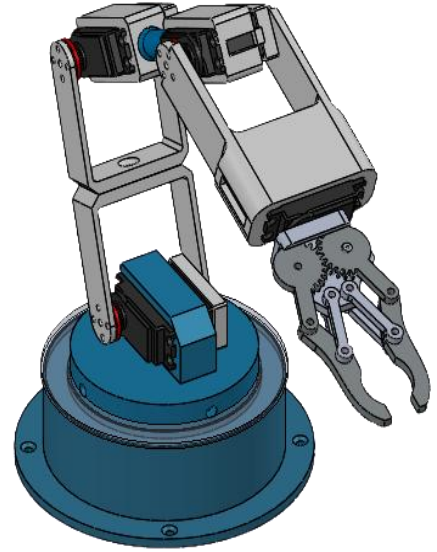


Figure 1: CAD Design

II. Mechatronics Hardware Design

A-Requirements

The proposed system required the design of a lightweight 6-DOF robotic arm capable of performing accurate and high-speed pick-and-place operations. The robotic arm was designed to minimize inertia, enabling smooth continuous motion and efficient operation using lower-torque actuators. At the same time, the mechanical structure needed to maintain sufficient rigidity and stability to withstand varying payloads without structural deformation. The primary design objectives were speed, precision, and mechanical robustness. In addition, the system was expected to provide reliable and repeatable motion control suitable for practical robotic manipulation tasks.

B- Research and Conceptual Design

We began the conceptual design phase by reviewing existing 6-DOF robotic arm designs available from online repositories such as GrabCAD, MakerWorld, and other open-source platforms. From this survey, two reference designs were selected as primary inspiration, as illustrated.



Figure 2

The first design offered advantages in terms of accessibility, ease of fabrication, and a compact overall structure; however, it exhibited relatively high material usage, particularly in the links, which increased weight and cost.



Figure 3

The second design demonstrated a more material-efficient structure but lacked available detailed models for direct implementation. To balance these trade-offs, the final design adopted the base structure of the first model and the link architecture of the second, combining manufacturability with material efficiency.

C- Initial Design and Fabrication

1) Base Design

The base design was initially adapted from an existing model (Model 1-a) and modified to meet the system requirements. The structure was scaled by 40% to improve stability and accommodate the overall arm dimensions. The servo mounting interface was redesigned to support an MG996 continuous rotation servo instead of the originally specified SG90. In addition, a dedicated opening was introduced for servo wiring to facilitate easier routing and maintenance, and a groove was integrated to securely house the servo horn within the upper base section.

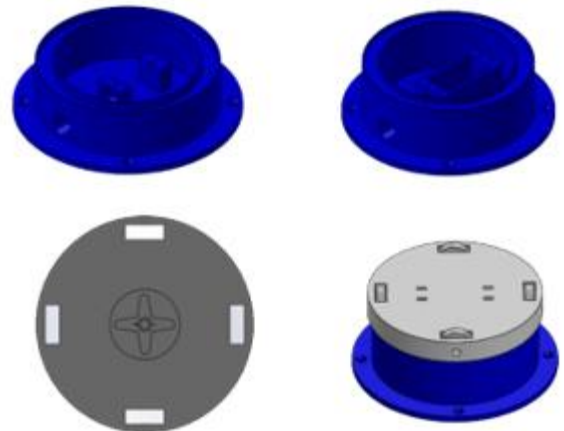


Figure 4: Main changes to the base

Following fabrication via 3D printing, several mechanical issues were identified, particularly clearance mismatches in the servo horn groove and instability in the bearing interface. The plastic-on-plastic contact between the rotating components resulted in unwanted slipping during operation. To resolve these issues, the bearing system was removed, the top component was resized to fit more securely within the lower section, and an acrylic reinforcement ring was added to improve alignment and structural stability.

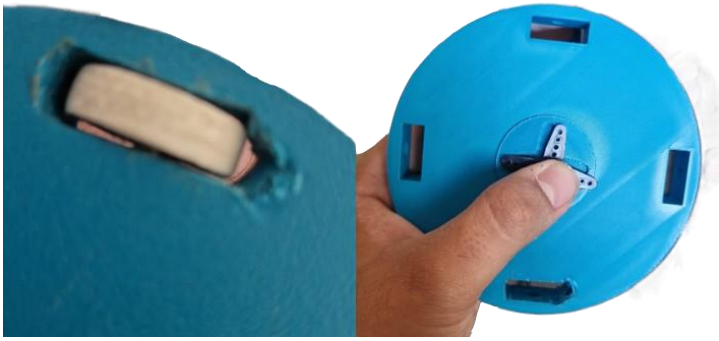


Figure 5: Defects in the clearance

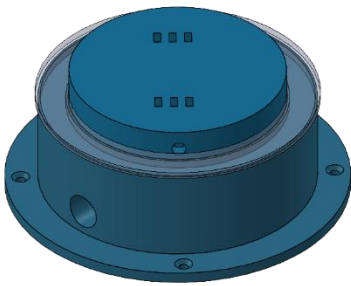


Figure 6: Final Base Design

2) Links

Link 1 was inspired by Model 1-b to reduce overall weight while maintaining structural performance. It consists of two H-shaped parts glued back-to-back. Initial tests showed poor edge contact, so a 45-degree chamfer was introduced to improve alignment. However, the structure failed under torsional load, so an M5 bolt with a washer and nut was added at the center to reinforce the link. This significantly improved rigidity, as shown in Fig. 7.

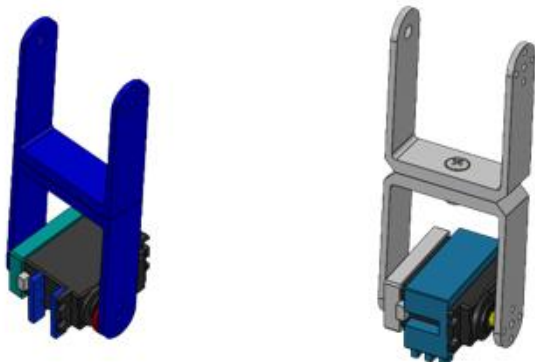


Figure 7: Initial & Final Link 1 Design

Link 2 consists of two separate components connected through a central rod. During initial testing, it was observed that the servo mounting grooves required extension to the full length of each part to ensure proper fit and alignment. To improve structural integrity, a key-slot mechanism was introduced on the central rod, with corresponding keys integrated into both link parts. This modification significantly increased resistance to torsional loads and improved overall mechanical stability.

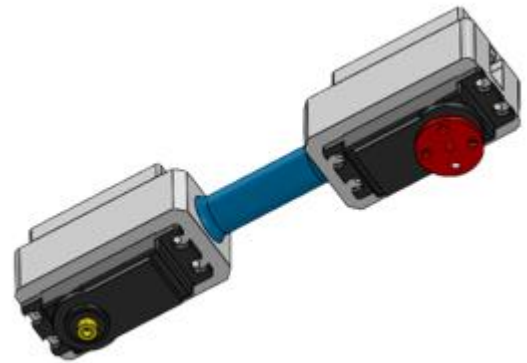


Figure 8: Link 2 Final Design

For Link 3, the design was thickened to transform the H-shaped part (link 1) into a robust housing for the end-effector servo. The gripper was transitioned to cast aluminum to provide a high strength-to-weight ratio and greater durability.

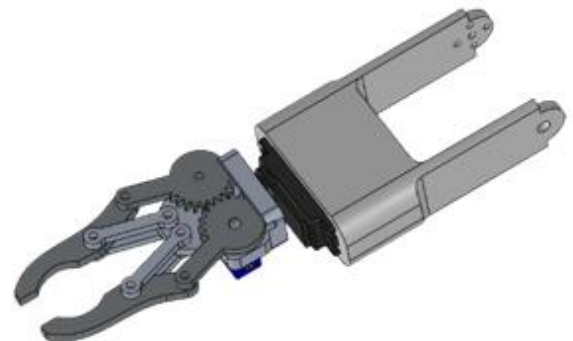


Figure 9: Link 3 & Gripper Final Design

D- Material Selection

PLA+ filament was selected for the base and linkages due to its lightweight nature, adequate mechanical strength, relatively low cost, and ease of fabrication using fused deposition modeling (FDM). It also demonstrates good impact resistance, reliable printability, and low moisture sensitivity, making it suitable for rapid prototyping and structural components in lightweight robotic systems [3], [4].

The gripper was manufactured from cast aluminum due to its high strength-to-weight ratio, as the end-effector is the most critical component of the system. Cast aluminum provides high structural integrity, improved wear resistance, and reliable performance under repeated loading cycles, making it suitable for repetitive pick-and-place operations where durability is essential [5].

E- Finite Element Analysis

A static finite element analysis (FEA) was performed in SOLIDWORKS Simulation to evaluate the structural performance of the assembly under a 2 kg payload applied at the gripper. The results indicate a minimum factor of safety of approximately 1.97, confirming that the design operates within acceptable structural limits. The displacement analysis shows a maximum deflection of 0.705 mm at the end effector tip under the applied load.

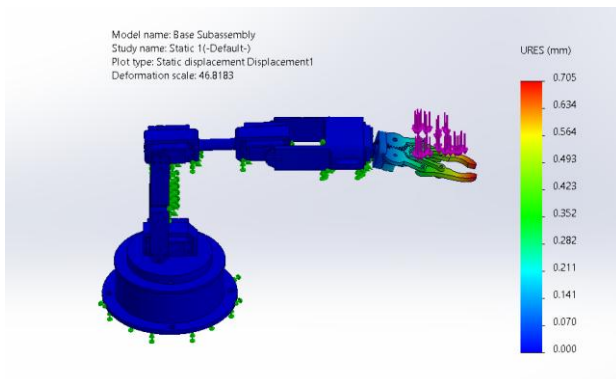


Figure 10: Displacement Diagram

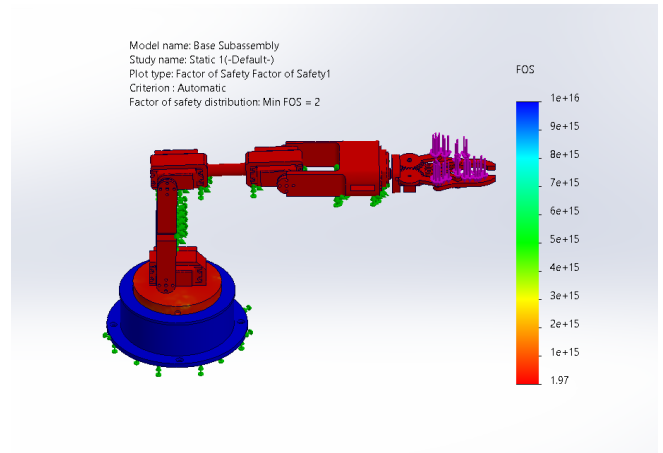


Figure 11: FOS Plot

F- Fabrication

The models were exported as STL files, and settings were adjusted in Bambu Lab's Bambu Studio software. The print infill was set to 40% with the fill pattern being gyroid (the strongest pattern) and 3 walls. The fabrication process took two steps:

- 1) Testing phase: parts such as the top base, link 1, and link 2 were printed with a lower infill of 15% on an Elegoo 3D printer, as shown in the figures below. This allowed us to check for clearance and fitment issues.
- 2) Final manufacturing stage: The final components were printed using a Bambu Lab A1 printer.



III. Electrification

The electrical architecture of the robotic arm was developed around the ESP32 microcontroller due to its compact size, low power consumption, integrated wireless communication capabilities, and compatibility with real-time robotic control applications. The ESP32 was responsible for handling actuator communication, receiving motion commands, and interfacing between the control software and the physical robotic hardware. The servo motors were electrically connected to the ESP32 through a distribution terminal block, which simplified wiring organization, improved connection reliability, and facilitated power and signal distribution across the system.

To enable modular and scalable robotic communication, the system was integrated with the Robot Operating System 2 (ROS2). ROS2 was selected due to its real-time communication support, distributed architecture, and extensive robotics development tools. The ROS2 framework was used to transmit motion commands, process control signals, and coordinate communication between the simulation environment and the robotic arm hardware.

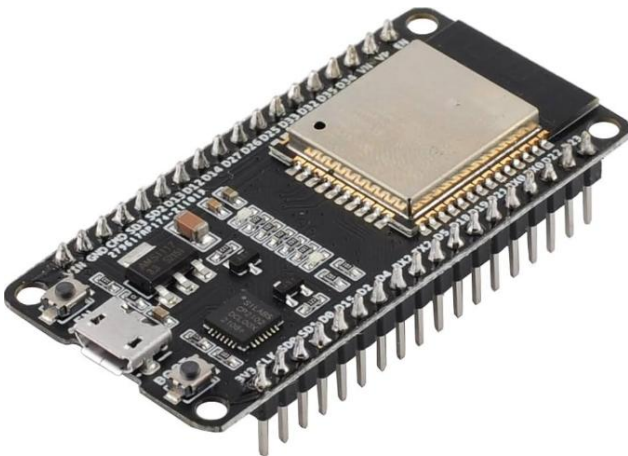


Figure 13: ESP32

The ESP32 was programmed to subscribe to ROS2 topics containing the desired joint commands generated from the control and trajectory planning modules. These commands were then converted into actuator control signals for the revolute joints of the manipulator. The system architecture allowed bidirectional communication between the ROS2 environment and the embedded controller, enabling continuous monitoring and command execution during operation.

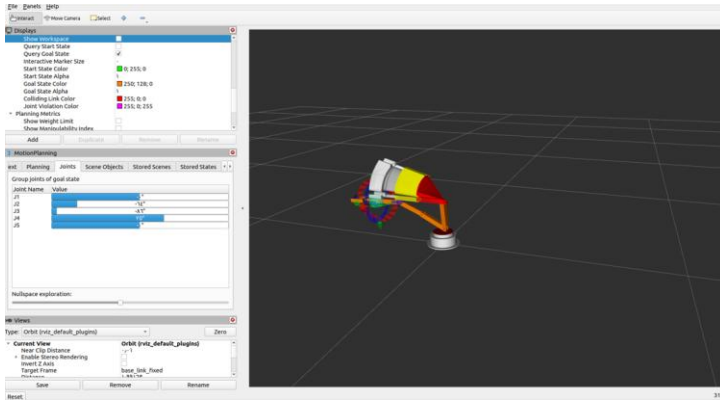
In addition, the ROS2-based implementation provided flexibility for future expansion, including integration with sensors, computer vision systems, and higher-level autonomous control algorithms. The combination of ESP32 and ROS2 established a lightweight and efficient hardware-software platform suitable for robotic manipulation and real-time motion control applications.

IV. ROS2 and MoveIt Software Architecture

The robotic arm control system was implemented using ROS2 in combination with MoveIt to provide motion planning, trajectory execution, and communication between the simulation environment and the embedded ESP32 controller. The software architecture was organized into multiple ROS2 nodes responsible for trajectory generation, inverse kinematics, motion execution, and joint-level communication. This modular framework enabled efficient integration between the robotic hardware and high-level planning algorithms.

The ROS2 program begins by initializing the communication environment and creating the required nodes for manipulator control. Publisher nodes are used to transmit desired joint trajectories and actuator commands,

while subscriber nodes receive feedback related to joint states and system monitoring. Communication between nodes is achieved through ROS2 topics carrying position, velocity, and trajectory data. MoveIt was integrated into the system to handle motion planning and inverse kinematics computations for the robotic manipulator. The MoveIt framework generates collision-aware trajectories and computes the required joint configurations needed to achieve the desired end-effector pose. This allowed smooth path planning for pick-and-place operations while maintaining stable and continuous manipulator motion. The generated trajectories were then transmitted through ROS2 topics to the ESP32 controller for physical execution.



Several callback and timer functions were implemented to continuously process incoming commands and synchronize actuator control during operation.

The ESP32 receives the joint trajectory commands generated by MoveIt and converts them into actuator control signals for the revolute joints of the robotic arm. Feedback data from the joints can additionally be transmitted back to the ROS2 environment for monitoring and debugging purposes.

The integration of ROS2, MoveIt, and the ESP32 controller established a scalable and flexible robotic control framework suitable for trajectory planning, manipulation tasks, and future expansion toward autonomous robotic applications.

V. MATLAB-Based Modeling and Simulation

A- Simulink Block Architecture

A comprehensive MATLAB/Simulink simulation framework was developed using Simscape Multibody to model and analyze the kinematic behavior of the proposed 6-DOF robotic arm. The simulation environment was designed to emulate the physical motion of the manipulator while enabling integration between trajectory generation, inverse kinematics, Jacobian-based analysis, and joint-level motion control within a unified architecture. The framework provides a physics-based environment for evaluating the robotic arm performance prior to fabrication and experimental implementation.

The robotic manipulator was first modeled using a rigid body representation to define the kinematic chain and spatial relationship between the links and revolute joints. This representation was connected to several robotics function blocks responsible for performing kinematic calculations throughout the simulation process. Among the primary functional blocks used were the 'Get Transform' and 'Get Jacobian' blocks. The 'Get Transform' block was implemented to compute the homogeneous transformation matrix between the robot base frame and the end-effector frame. This transformation matrix provides the position and orientation of the end effector relative to the base coordinate system and allows continuous monitoring of the manipulator pose during trajectory execution.

To supply the required inputs to the transformation block, the joint variables were combined using a MUX block that organized the angular joint data into a suitable vector form compatible with the robotics toolbox functions. This arrangement simplified the signal flow and enabled efficient matrix-based computations throughout the simulation environment.

In addition to pose estimation, manipulator velocity analysis was performed using the *Get Jacobian* block. This block generates the manipulator Jacobian matrix of dimensions $6 \times N$, where N corresponds to the number of revolute joints in the robotic system. The Jacobian matrix establishes the relationship between joint-space velocities and Cartesian-space linear and angular velocities of the end effector. The Jacobian analysis was essential for evaluating the motion characteristics of the manipulator and understanding how variations in joint motion affect the velocity of the end effector during operation.

Inverse kinematics was implemented using the Inverse Kinematics functional block provided within the Robotics System Toolbox. This block calculates the required joint angles needed to achieve a desired end-effector pose in Cartesian space. The inverse kinematics solver contains three main terminals: pose, weights, and initial guess. The pose terminal defines the target position and orientation of the end effector, which can either be manually specified or generated dynamically through trajectory planning blocks. The weights terminal allows prioritization between orientation accuracy and positional accuracy during the optimization process, enabling flexible control depending on the task requirements. The initial guess terminal provides an estimated starting configuration for the solver, improving convergence speed and minimizing the likelihood of unstable or nonphysical solutions.

For continuous motion generation, a polynomial trajectory block was integrated into the simulation environment. This block updates the desired end-effector position at every simulation cycle, generating smooth position references suitable for pick-and-place operations.

Polynomial trajectories were selected due to their ability to produce continuous velocity and acceleration profiles, reducing sudden motion discontinuities that may negatively affect manipulator stability and actuator performance. The generated Cartesian trajectory was then passed directly to the inverse kinematics solver to obtain the corresponding joint-space commands.

The physical structure of the robotic arm was represented through a subsystem block generated automatically after exporting the CAD assembly into Simscape Multibody. This subsystem contains all revolute joints, rigid bodies, coordinate frames, and mechanical constraints required to simulate the actual mechanical behavior of the manipulator. To improve readability and simplify system integration, customized input and output ports were added to the subsystem architecture. The subsystem accepts joint angle commands as inputs while providing angular position and angular velocity measurements as outputs for each revolute joint within the system.

To further analyze manipulator motion, the angular velocities obtained from the subsystem outputs were combined using a 5×1 MUX block. These combined signals were subsequently multiplied by the Jacobian matrix to compute the corresponding Cartesian velocity of the end effector. This process enabled evaluation of velocity propagation across the manipulator chain and provided additional insight into the relationship between joint-level motion and end-effector dynamic behavior.

Several display and monitoring blocks were additionally integrated throughout the simulation architecture for debugging and validation purposes.

These blocks were used to visualize the real-time end-effector position, monitor joint responses, and verify the correctness of the inverse kinematics solutions during operation. The displays also facilitated identification of singularities, unexpected trajectory deviations, and unstable joint configurations throughout the simulation process.

The developed Simulink framework successfully provided a complete virtual environment for evaluating robotic arm kinematics, trajectory generation, and manipulator motion behavior prior to physical implementation. The integration of Simscape Multibody with inverse kinematics and Jacobian-based analysis enabled accurate simulation of the robotic arm and established a reliable foundation for future control and experimental studies.

VI. Discussion and Conclusion

In this work, the design, development, simulation, and control implementation of a 6-DOF robotic manipulator for pick-and-place applications were presented. The mechanical structure was optimized to achieve a balance between lightweight construction, structural rigidity, and cost efficiency through the combined use of PLA+ and cast aluminum components.

MATLAB/Simulink and Simscape Multibody were utilized to model the manipulator and evaluate its kinematic behavior using inverse kinematics, Jacobian analysis, and trajectory generation techniques. In addition, ROS2 and MoveIt were integrated with the ESP32 microcontroller to establish a scalable framework for trajectory planning and real-time robotic control.

Structural validation through finite element analysis demonstrated that the manipulator could safely withstand the required payload while maintaining acceptable displacement and factor of safety values.

Experimental and simulation results verified the feasibility of the proposed system and confirmed the effectiveness of the developed control and communication architecture for robotic manipulation tasks.

Although the developed system achieved the intended functionality, several limitations remain. Factors such as servo backlash, structural flexibility in 3D-printed components, and actuator precision may affect trajectory accuracy during continuous operation. Future work will focus on improving control accuracy through closed-loop feedback systems, integrating computer vision for autonomous object detection, and implementing advanced motion planning and dynamic control algorithms to enhance the overall performance and reliability of the robotic manipulator.

VII. References

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